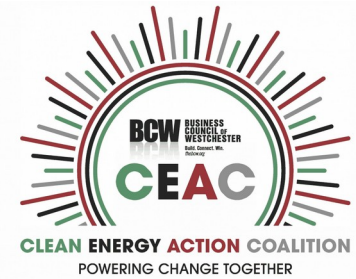


PASSIVHAUS FOR MIXED USE & MULTIFAMILY

Breakout Session Summary

Westchester Clean Energy Action Conference



This breakout session, led by members of the Phuis Alliance New York Southern Hudson Valley affiliate, made the case for Passive House as a practical, beautiful, and increasingly affordable standard, not just a niche for single-family homes. Through built examples ranging from historic landmarks to affordable multi-family housing, the panel walked through the building science, the comfort and health benefits, and the economics that determine whether projects actually get built to the standard.

SESSION PARTICIPANTS

Moderator  **Christina Griffin** Principal
CGA Studio Architects



Keiko & Bill Spade Principals | Sasaki+Spade Architects | William
R Spade Architecture & Development Consulting



Rob Hothan MSc Energy Management &
Future Building Solutions | DUK Krems

1. WHAT PASSIVE HOUSE ACTUALLY IS

The panel grounded the conversation in five principles: continuous insulation, an airtight envelope, no thermal bridges (the “thermal highways” in ordinary walls), high-performance triple-pane windows that are properly oriented for sun and shade, and balanced mechanical ventilation through an HRV or ERV. Added together, a Passive House needs roughly 90% less energy than a code-built building, with the small remaining load supplied by sunlight, appliances, and body heat. The European benchmark cited was about 15 kWh per square meter per year for heating and cooling. The presenters stressed that the standard is “simple, but not easy”: the principles are clear, but execution demands genuine building-science discipline.

2. SELL COMFORT, NOT JUST ENERGY

A recurring argument was that because energy is still relatively cheap in the US, the most persuasive case for Passive House is comfort, not utility savings. Drawing on P.O. Fanger’s comfort research, Robert Hothan explained why a person feels cold next to a double-pane window even in a heated room: the cold glass surface radiantly pulls heat from the body, and chilled air sliding down the glass creates a convection draft that feels like a leak. Triple-pane Passive House glazing keeps interior surfaces warm (around 60°F on a 0°F day), eliminating both the radiant loss and the draft, and that even, quiet, comfortable interior is what sells the standard.

3. HEALTH AND WELLNESS

Beyond comfort, the panel framed Passive House as a health product. Because the envelope is so airtight, balanced ventilation (ERV in this humid climate) continuously supplies tempered, filtered fresh air while recovering up to ~90% of the energy in the exhaust. All-electric systems mean no combustion and no off-gassing indoors. Crucially, the assemblies are designed to be vapor-permeable so walls can dry in both directions across the seasons, which, combined with airtightness, prevents the mold and pest problems common in conventional construction.

“We maintain fresh air. That’s the wellness aspect, that we don’t have mold issues or pest issues in the interior of the house.” - **Keiko Sasaki, Sasaki+Spade Architects**

4. CODE IS THE FLOOR, NOT THE GOAL

A theme the architects returned to repeatedly: building code is the minimum, the least you are legally permitted to build, and meeting it alone guarantees substantial energy waste. The team pitches Passive House methods to every client, and even when a client won't commit to full certification, they push the envelope, insulation, and all-electric systems above code. The corollary is reuse: existing buildings hold enormous embodied energy, so renovating rather than demolishing is one of the most sustainable moves available.

"The code is the least acceptable building that you can build, the worst building that you're legally allowed to build."

Bill Spade, Sasaki+Spade Architects

5. RENOVATION AND HISTORIC BUILDINGS

Much of the session showcased retrofits, proving the standard is not limited to new construction. Examples included a landmarked 1891 building in Yonkers fitted with custom historic-match triple-pane windows, a former 1885 Quaker meeting house in Chappaqua rebuilt as two affordable units to the EnerPHit renovation standard using volunteer Habitat/Fuller Center labor, and German retrofits where prefabricated panels are craned onto occupied buildings, even wrapping former concrete balconies into the thermal envelope to kill a massive thermal bridge. The consistent message: double-wall or Larson-truss assemblies and continuous air barriers can bring even very old buildings to the standard while preserving their character.

"The most important thing is the tremendous amount of embodied energy in an existing building. You've got to keep it."

Robert Hothan, Building Scientist

6. THE ECONOMICS AND SCALING UP

On cost, the panel distinguished sharply between building types. Per PHIUS data, the incremental cost of Passive House for multi-family is only about 1–2% over a code-compliant baseline, versus roughly 8% for single-family, because larger buildings have a far better surface-area-to-volume ratio, more shared "adiabatic" walls, and downsized mechanical systems. The persistent obstacle is the windows: fear of unfamiliar (non-Marvin) brands and upfront cost is the most common reason builders fall back to code, especially after years of NIMBY-driven approval delays drain budgets. Prefab and panelized systems (e.g., Bensonwood) and subsidies (e.g., a state net-zero/Passive House homeownership incentive) were offered as ways to de-risk and close the gap.

"The bigger the project, the more the costs go down."

Christina Griffin, CGA Studio Architects

KEY TAKEAWAYS

1

Passive House rests on five principles:

1. Continuous insulation.
2. An airtight envelope.
3. No thermal bridges.
4. High-performance triple-pane windows with proper orientation.
5. An ERV/HRV.

Together they cut energy use by roughly 90%.

2

Sell comfort, not just savings. With energy still cheap in the US, the winning argument is warm interior surfaces, no cold-window drafts, and even, quiet temperatures.

3

It is also a health product: balanced ventilation delivers filtered fresh air, all-electric means no combustion or off-gassing, and airtight yet vapor-permeable walls prevent mold and pests.

4

Code is the floor, not the goal. It is the lowest quality building you are legally allowed to build; aim above it even when clients won't commit to full certification.

5

Passive House works on old and historic buildings via the EnerPHit standard, and reusing existing structures preserves embodied carbon, sometimes even wrapping balconies and facades to remove thermal bridges.

6

Economics favor scale: incremental cost is only about 1–2% over code for multi-family versus about 8% for single-family, thanks to surface-to-volume ratio, shared walls, and smaller mechanical systems.

7

The windows are the Achilles' heel. Fear of unfamiliar brands and upfront cost most often pushes projects back to code; prefab/panelized systems and net-zero subsidies help close the gap.